

# Madhur Chavan

Pune, India | +91-7887904958 | [chavanmadhur65@gmail.com](mailto:chavanmadhur65@gmail.com) | [Linkedin](#) | [Github](#)

## EDUCATION

### MIT World Peace University

B.Tech in Computer Science and Engineering

Pune, India

Aug. 2023 – May 2026

### Government Polytechnic Pune

Diploma in Computer Engineering

Pune, India

Aug. 2020 – May 2023

## EXPERIENCE

### SDE Intern

LTI Mindtree

Sept. 2025 – Dec. 2025

Pune, India

- Spearheading the migration of the search and analytics platform from Elasticsearch to **OpenSearch**, enhancing system scalability and eliminating vendor lock-in.
- Customizing a Java agent to generate **custom traces**, enabling deep monitoring of microservices with tools like **Jaeger** and **Zipkin** and improving root cause analysis time.
- Collaborating with the observability team to build and maintain real-time monitoring dashboards using **Kibana** and OpenSearch Dashboards for performance visualization.

### SDE Intern

Deavnet Solutions

July 2022 – Aug. 2022

Pune, India (Remote)

- Engineered and deployed backend RESTful APIs using ASP.NET for enterprise applications, reducing response time by 15% and supporting over 10K daily transactions.
- Designed and normalized SQL Server database schemas, improving query performance by 25% while ensuring security for 100K+ user records.
- Collaborated with a cross-functional team to resolve over 30 software bugs, improving release stability by 20% and boosting customer satisfaction scores.

## PROJECTS

### Natural Language to SQL Conversion System | Python, Flask, T5 Transformer, Scikit-learn, MySQL, React

- Built a hybrid NLP application to convert natural language into SQL queries, achieving **92% accuracy** on a custom dataset of over **2,000 queries**.
- Developed an end-to-end system with a React frontend, Flask backend, and a T5-based query generator capable of handling **real-time queries under 500ms**.
- Enabled intuitive database access for **non-technical users**, reducing manual SQL writing effort by **70%** and enhancing internal data accessibility.

### Gesture Controlled Computer System | Python, OpenCV, MediaPipe, Tkinter

- Engineered a real-time gesture recognition system using OpenCV and MediaPipe, achieving **90% detection accuracy** and sub-**1s response time**.
- Designed a gesture vocabulary of **10+ actions** to automate tasks like scrolling, clicking, and navigation, boosting user productivity by **40%**.
- Created a Tkinter-based GUI enabling configuration and providing **instant visual feedback**, improving user interaction and ease of use.

### Sign Language Detection System | Python, OpenCV, MediaPipe, TensorFlow

- Developed a real-time sign language recognition system using TensorFlow and OpenCV, translating hand gestures to text with **93% model accuracy**.
- Trained on a dataset of **5,000+ labeled gesture frames** and deployed the model to support **live video translation at 15 FPS**, aiding accessibility for the hearing-impaired.

## TECHNICAL SKILLS

**Languages:** C, C++, Java, Python, SQL, JavaScript, C#, PHP

**Frameworks:** Flask, ASP.NET, React, Tkinter

**AI & Data Science:** TensorFlow, Scikit-learn, OpenCV, MediaPipe

**Developer Tools:** Git, GitHub, Docker, Postman, Visual Studio, MySQL Workbench

**Hardware & Embedded:** ESP32, Microcontrollers, MicroPython